

# MPMC LAB - Experiment 5

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## AIM

To develop, compile and simulate programs for arithmetic operations and waveform generation using ARM7 processor.

## ALGORITHM - Arithmetic Operations

Addition: Load values → Add → Store result.

Subtraction: Load values → Subtract → Store result.

Multiplication: Load values → Multiply → Store result.

Division: Repeated subtraction method to get quotient.

## CODE - Arithmetic Operations

```
MOV R0, #04
MOV R1, #02
ADD R2, R0, R1
SUB R3, R0, R1
MUL R4, R0, R1
```

## Sawtooth Wave

```
MOV R0, #0
LOOP: STR R0, [R1]
ADD R0, R0, #1
CMP R0, #255
BNE LOOP
```

## Triangle Wave

```
UP: ADD R0, R0, #1
CMP R0, #255
BNE UP
DOWN: SUB R0, R0, #1
CMP R0, #0
BNE DOWN
```

## Square Wave

```
MOV R0, #0xFF
STR R0, [R1]
BL DELAY
MOV R0, #0x00
STR R0, [R1]
```

## Buzzer Code

```
START:  
MOV R0, #1  
STR R0, [R1]  
BL DELAY  
MOV R0, #0  
STR R0, [R1]  
BL DELAY
```

## RESULT

The programs were successfully executed and verified.